DATA ITEM DESCRIPTION

Title: AIRCRAFT STORE LOADS DESIGN CRITERIA REPORT

Number: DI-SESS-82254A
AMSC Number: N10471
DTIC Applicable: N/A
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20240408
Limitation: None
GIDEP Applicable: No
Project Number: SESS-2024-005

Use/relationship: This report describes the aircraft store loads criteria that will be used in the structural design of a store carried on an aircraft for all store variants.

This Data Item Description (DID) contains the format, content, and intended use information for the data product resulting from the work task described by the contract SOW.

Requirements:

1. Reference documents. None

2. Format. Contractor format acceptable.

3. Content. The Aircraft Store Loads Design Criteria Report shall:

3.1. The following criteria & documentation requirements apply to all store variants.

3.2. Loads documentation for all store variants shall include:

3.2.1. All design weights as specified by the contract;

3.2.2. Center of gravity (CG) envelopes including weight CG diagrams and the derivations;

3.2.3. Design load factor and airspeed diagrams (V-n diagrams) for symmetrical and unsymmetrical flight;

3.2.4. Design load factors, airspeeds, and rotational velocities and accelerations;

3.2.5. Elevated temperature criteria;

3.2.6. Internal store pressure criteria;

3.2.7. Design landing and ground handling operations;

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3.2.8. Design symmetric and asymmetric maneuvers, including steady state and dynamic maneuvers, resulting in high dynamic load amplification (i.e. buffet);

3.2.9. Spanwise load factor & V-M-T (Shear-Moment-Torsion) distributions along pertinent design axes for store (as stipulated by contract);

3.2.10. Six Degree of Freedom (6DOF) store center of gravity (cg) translational and rotational design accelerations (as stipulated by contract);

3.2.11. Store-to-rack interface definitions and loads including all swaybraces and hooks (as stipulated by contract);

3.2.12. Design load and/or actuation load requirements associated with door, panel, and/or turbine operation of radiating stores;

3.2.13. Store-to-store and store-to-aircraft clearances;

3.2.14. Store handling, cradling, transport, and abuse criteria;

3.2.15. All critical loading configurations as specified by the contract.

3.3. Define the specific design conditions selected for analysis including:

3.3.1. A list of store loading configurations selected for analysis, together with the basis for selection of those conditions and configurations;

3.3.2. A discussion of design features, which affect determination of critical conditions;

3.3.3. All sources of aerodynamic and inertial data;

3.3.4. All methods and assumptions to be used in the calculation of aerodynamic and inertial loads.

3.4. Define the structural capability by superimposing all limiting factors, including control power, angle of attack, and buffet, on altitude vs Mach number envelopes.

3.5. Define the effects of weight, CG, and variable geometry component positions;

3.6. Document a detailed description of the control system, its components, and their functions.

3.7. Document the sources and complete explanations of all loads on the system and its components.

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3.8. Define the following data for field landings, arrested landings, Field Carrier Landing Practice landings, field takeoffs, catapults, taxiing, and ground handling:

3.8.1. A summary of all in flight loading conditions that will be combined with inertial loads;

3.8.2. The catapult's horizontal accelerations used for each type of catapult;

3.8.3. The arresting hook forces and horizontal accelerations used for each type of arresting gear;

3.8.4. The range of landing gear variation in servicing that will be used for design;

3.8.5. A summary of all ground handling, taxi, securing, and miscellaneous ground loading conditions;

3.8.6. Substantiating data, pertinent references, and amplifying data showing that the derivations of the required range of parameters comply with contract requirements;

3.8.7. Drawings showing required attitudes with auxiliary doors in the fully extended and fully closed position relative to the store reference line;

3.8.8. The clearance between the adjacent store or structure.

End of DI-SESS-82254A

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